

Rezaul Karim, Ph.D.

Artificial Intelligence / Machine Learning

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Research Statement Summary

I am a machine learning researcher whose program centers on efficient, scalable foundation-model training and memory-augmented agentic AI. My PhD and subsequent industry-research career have produced peer-reviewed contributions at CVPR, EMNLP, ICRA, and WACV, one granted patent and two pending applications, spanning multiscale video transformers, efficient LLM training/caching, and reinforcement learning for tool-use agents. As an academic researcher, I would build a research program bridging efficient large-scale training, agentic and memory-augmented AI systems, and their translation into deployable, production-grade methods, while contributing teaching across data science, machine learning, and systems curricula.

Education

Ph.D., Electrical Engineering & Computer Science — York University <i>Thesis: Spatial-Temporal Attention Models for Video Understanding</i>	<i>Graduated 2024</i>
M.Sc., Computer Science — University of Manitoba <i>Thesis: Gated Architectures for Image Segmentation</i>	<i>Graduated 2019</i>
B.Sc., Computer Science & Engineering — BUET	<i>Graduated 2015</i>

Academic & Industry Research Appointments

Senior Researcher, Huawei Canada Corp.	<i>2024 – Present</i>
- Led experimental studies translating research ideas into deployable proof-of-concepts on production-scale clusters for foundation model training, inference, and agentic AI systems. - Conduct applied research on efficient architectures, scalable training strategies, and memory-augmented agent systems for large language and multimodal foundation models. - Research reinforcement learning and reward function design for tool-use agents, including speculative decoding techniques for efficient reinforcement learning. - Develop memory-augmented agent architectures, including procedural memory and skill-library design for long-horizon skill acquisition. - Developed efficient pretraining and post-training algorithms improving convergence stability, model efficiency, and deployment performance. - Proposed and validated zero-overhead tensor parallelism as a novel research contribution (patent submitted).	
Researcher (Intern), Noah's Ark Lab, Huawei Canada Corp.	<i>2022</i>
Conducted research on vehicle trajectory prediction for autonomous driving, with emphasis on efficiency and robustness.	
Graduate Research Assistant, Vision Lab, York University	<i>2019 – 2024</i>
- Developed efficient spatial-temporal attention mechanisms and a multiscale video transformer. - Proposed MED-VT, a novel multiscale encoder-decoder video transformer achieving state-of-the-art performance on video segmentation (CVPR 2023). - Conducted large-scale experimental evaluations with systematic ablation and efficiency analyses of video transformers for vision and multimodal segmentation tasks.	
Graduate Research Assistant, Vision Lab, University of Manitoba	<i>2017 – 2019</i>
- Introduced feedback and iterative gating architectures for efficient semantic segmentation (WACV). - Studied adversarial robustness and efficiency trade-offs in deep neural networks.	

Teaching Experience

Teaching Assistant, Department of EECS, York University	<i>2019 – 2024</i>
EECS 3101: Design and Analysis of Algorithms (Section E).	
Teaching Assistant, Department of Computer Science, University of Manitoba	<i>2017 – 2019</i>
- Programming Practices; Data Structures and Algorithms. - Grader: Introductory Computer Science; Introduction to Computer Systems; Object Orientation.	

Selected Publications

1. Rezaul Karim, Maryam Dialameh, Walid Ahmed, Yang Liu, and Boxing Chen. EPAS: Efficient Training with Progressive Activation Sharing. *Canadian AI*, 2026 (accepted).
2. Dialameh, Maryam, Rezaul Karim, Hossein Rajabzadeh, Omar Mohamed Awad, Hyock Ju Kwon, Boxing Chen, Walid Ahmed, and Yang Liu. ECHO-LLaMA: Efficient Caching for High-Performance LLaMA Training. *EMNLP*, 2025.
3. Rezaul Karim, He Zhao, Richard P. Wildes, and Mennatullah Siam. MED-VT: Multiscale Encoder-Decoder Video Transformer with Application to Object Segmentation. *CVPR*, 2023.
4. Rezaul Karim, M. A. Islam, and N. Bruce. Distributed Iterative Gating Networks for Semantic Segmentation. *WACV*, 2020.
5. Rezaul Karim, M. A. Islam, and N. Bruce. Recurrent Iterative Gating Networks for Semantic Segmentation. *WACV*, 2019.

Additional publications in CRV, SPIRE, IEEE SSCI, and Nature Scientific Reports.

Selected Patents

- Method and System for Predicting a Trajectory — US Patent Application 18/485672 (2025).
- Zero-Overhead Tensor Parallelism for Scalable LLM Training — Patent Submitted (2025).

Grants, Honors & Awards

- DRI-EDIA Champions Pilot Program — Digital Research Alliance of Canada (2024).
- Multiple graduate scholarships and fellowships, including VISTA, LGES, MGS, and IGSES.

Academic & Professional Service

- Conference Reviewer: NeurIPS, ACL, CVPR, WACV, IEEE RA-L.
- Journal Reviewer: Computer Vision and Image Understanding (CVIU).

Technical Expertise

Languages & Frameworks: Python, PyTorch, TensorFlow.

LLM Systems: Pre/post-training, alignment, agent workflows, tool use, memory systems, text-centric AI/ML.

Foundation Models: Transformers, vision, multimodal models, LLMs.

Agentic AI: Reinforcement learning, reward function design, tool-use agents, memory-augmented agents, procedural memory, skill libraries.

Distributed AI: Parallelism, communication optimization, scalable training and inference.